PHASE 0 · RESEARCH, DATA SOURCE AND PROPOSAL

RiderLink

Order Tracking and Rider Assignment for Food Delivery in Kigali

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github.com/Nshutihonore/honore_29683_riderLink

BACKGROUND AND PROBLEM

Nothing stops a rider from being overloaded

Riders

Delivery in Kigali runs on motorbike riders. Research links time pressure to unsafe riding and unstable loads.

Customers

When a dispatcher assigns orders by hand, one rider can hold too many, and food arrives late.

Restaurants

Platforms such as Vuba Vuba have grown since 2020, so order volume keeps rising.

MOTIVATION

A safety rule belongs in the database, not in a chat group

A real local problem

Riders in Kigali carry orders that dispatchers assign by hand.

A rule PL/SQL enforces

A trigger refuses an assignment the moment a rider is full.

A tool people can use

An Oracle APEX board lets dispatchers see load and assign orders.

OBJECTIVES

One main objective, four specific ones

Main: build an Oracle database with PL/SQL that records food-delivery orders and rider assignments and prevents over-capacity assignments.

1. Model the order process and normalize real data to 3NF.

2. Build procedures, functions, cursors and a package for orders and assignments.

3. Enforce the capacity rule with triggers and log every denied action.

4. Provide an Oracle APEX dispatch app that calls the procedures.

RESEARCH

Existing work predicts delivery time, not rider load

Source	What it shows	Gap for RiderLink
Garg et al. (2025); Yalçınkaya and Areta Hızıroğlu (2024)	Machine learning predicts delivery time	No rule limits orders per rider
Christie and Ward (2023)	Gig riders face more pressure and collisions	No system control on workload
Vuba Vuba (Wearetech Africa, 2025)	Leading local platform in Rwanda	Dispatch rules are not public

DATA SOURCE

Real delivery records from Kaggle

Food Delivery Dataset

Gaurav Malik, Kaggle

About 45,000 records
[confirm count]

Licence: [confirm educational use]

Dataset field	Becomes
Delivery person ID	rider
Restaurant coordinates	restaurant
Type of vehicle	vehicle_type
Weather, traffic	lookup tables
Order and pickup time	delivery_order

PROPOSED SOLUTION AND INNOVATION

A database rule, used through an APEX app

Solution

An Oracle database records restaurants, orders and riders. A trigger blocks assigning an order to a rider who is already at capacity.

Innovation: user interface

Tool: Oracle APEX. A dispatch board shows each rider’s load. An assign form calls assign_rider, and an audit view lists denied assignments.

PLANNED DATABASE

Seven tables that link riders to orders

vehicle_type
Holds the order limit

rider
Has one vehicle type

restaurant
Has many orders

lookup tables
Weather and traffic

delivery_order
One restaurant, many conditions

order_assignment
Links a rider to an order

audit_log
Allowed and denied actions

Third Normal Form

PL/SQL APPROACH

Each feature enforces or supports the capacity rule

Feature	Where it is used
Function get_active_load	Counts a rider’s open orders
Procedure assign_rider	Assigns a rider, raises an error when full
Procedure complete_delivery	Closes an assignment, frees capacity
Cursor	Loops over riders for the workload report
Package dispatch_pkg	Groups procedures, functions and exceptions
Trigger	Blocks a rider already at capacity
Audit trigger	Logs each allowed and denied action

EXPECTED RESULTS

What I will demonstrate

1

Real delivery data loaded into 3NF tables.

2

A full rider refused another order, with the attempt logged.

3

A live APEX dispatch board assigning riders.

FUTURE IMPROVEMENTS AND LIMITATIONS

What this version does not yet cover

Limitations

- Data comes from Indian cities, not Kigali
- Capacity per vehicle type is a design choice
- No live GPS tracking

Future improvements

- Load real Kigali order data
- Predict delivery time with machine learning
- Send rider alerts on a schedule

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